

Standard Operating Procedures — Production Line A

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1. Station Dependencies

ST01_FILLING → ST02_SEALING → ST03_LABELLING → ST04_PACKAGING. A fault at any station propagates downstream unless the line is halted.

Station ID	Zone	Depends On	Downstream Effect
ST01_FILLING	Production Area	—	Feeds ST02_SEALING
ST02_SEALING	Production Area	ST01	Feeds ST03; thermal fault → ST04 speed reduction
ST03_LABELLING	Production Area	ST02	Feeds ST04_PACKAGING
ST04_PACKAGING	Production Area	ST03	Final output
SRV01_SERVERROOM	Server Room	HVAC	SCADA control; failure → line halt
WRH01_WAREHOUSE	General Warehouse	COMPRESSOR	Cold storage — product quality
CHM01_CHEMICALSTORAGE	Chemical Storage	HVAC	Hazardous materials — safety critical
RND01_RDLAB	R&D Lab	HVAC	Precision experiments — climate sensitive
CAF01_CAFETERIA	Cafeteria	HVAC	Occupant comfort — food safety

2. Station-Level Operating Rules

2.1 ST01_FILLING

- RULE-ST01-01: FLW MUST be maintained between 112 and 128 L/min.
- RULE-ST01-02: If TMP exceeds 26°C, reduce speed by 10% and notify QC.
- RULE-ST01-03: PRS drop below 3.3 bar — inspect inlet valve within 5 min.

2.2 ST02_SEALING

- RULE-ST02-01: TMP MUST remain between 175°C and 195°C.
- RULE-ST02-02: TMP > 210°C (CRITICAL) — emergency stop ST02, ST03, ST04.
- RULE-ST02-03: CUR increase >1.5 A above nominal sustained >5 min — schedule maintenance.
- RULE-ST02-04: CUR drift at ST02_SEALING CORRELATES with speed reduction at ST04_PACKAGING due to shared electrical coupling. Monitor ST04_PACKAGING-SPD whenever ST02_SEALING-CUR triggers a WARNING.

2.3 ST03_LABELLING

- RULE-ST03-01: TEN MUST be between 1.8 and 2.6 N.
- RULE-ST03-02: CNT below 110 pcs/min — check for label jam. Below 100 (CRITICAL): halt.
- RULE-ST03-03: TEN STUCK >10 consecutive samples — sensor failure.

2.4 ST04_PACKAGING

- RULE-ST04-01: VIB > 0.20 mm/s (WARNING) — schedule maintenance within 24 h.
- RULE-ST04-02: VIB > 0.35 mm/s (CRITICAL) — emergency stop; inspect bearings.
- RULE-ST04-03: SPD drops below 1.00 m/s without operator command — verify RULE-ST02-04.

2.5 SRV01_SERVERROOM

RULE-SRV01-01: TMP MUST NOT exceed 25°C.

RULE-SRV01-02: TMP > 28°C (CRITICAL) — automatic controlled shutdown of all PLC nodes.

2.6 WRH01_WAREHOUSE (General Warehouse)

RULE-WRH01-01: TMP MUST remain between 0.5°C and 8°C. CRITICAL above 12°C — notify QC and quarantine batch.

RULE-WRH01-02: HUM above 85%RH — inspect door seals and activate dehumidifier.

2.7 CHM01_CHEMICALSTORAGE (Chemical Storage)

RULE-CHM01-01: TMP MUST remain between 10°C and 20°C. CRITICAL above 25°C — activate emergency ventilation and notify safety officer immediately.

RULE-CHM01-02: HUM MUST NOT exceed 50%RH. Elevated humidity accelerates degradation of hygroscopic reagents. Activate desiccant system if HUM >45%RH.

RULE-CHM01-03: Only personnel with hazmat training (technician/supervisor/manager role) are authorised to enter Chemical Storage. UNAUTHORIZED_ACCESS must be escalated within 2 minutes.

2.8 RND01_RDLAB (R&D; Lab)

RULE-RND01-01: TMP MUST remain between 18°C and 26°C for experimental validity.

RULE-RND01-02: HUM drift above 65%RH — activate dehumidifier. Above 75%RH (CRITICAL) — suspend sensitive experiments and inspect HVAC system.

2.9 CAF01_CAFETERIA

RULE-CAF01-01: TMP above 26°C — check ventilation system. Occupant comfort is the primary criterion.

RULE-CAF01-02: HUM above 70%RH — inspect drainage and activate ventilation.